



Team II

Development and Implementation of the BaSyx Viewer Plugin Extensions

22.05.2026

AGENDA



OUR TEAM / INTRODUCTION

PRODUCT OVERVIEW

REQUIREMENTS

TESTS

DEMO

DOCUMENTS/PM

OUR TEAM



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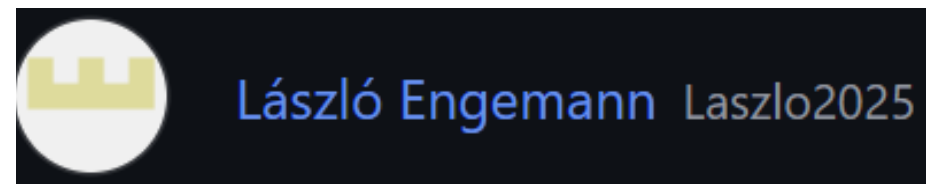
OUR TEAM



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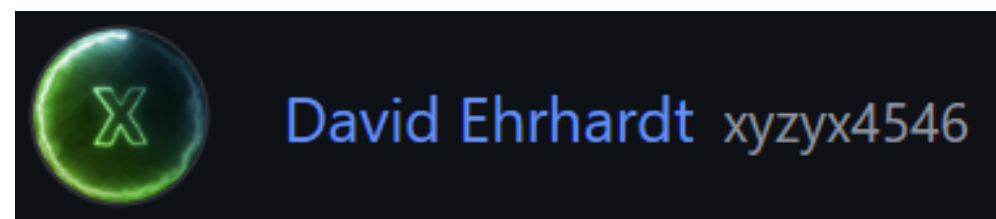
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Introduction



- Improved the Eclipse BaSyx AAS Web UI
- 2: Implemented sorting and filtering for shells
- 3: Added recursive search for AAS data
- 4: Integrated the Digital Nameplate generator
- 5: Extended metadata support for `createdAt` and `updatedAt`
- 6: Improved boolean labeling in the UI

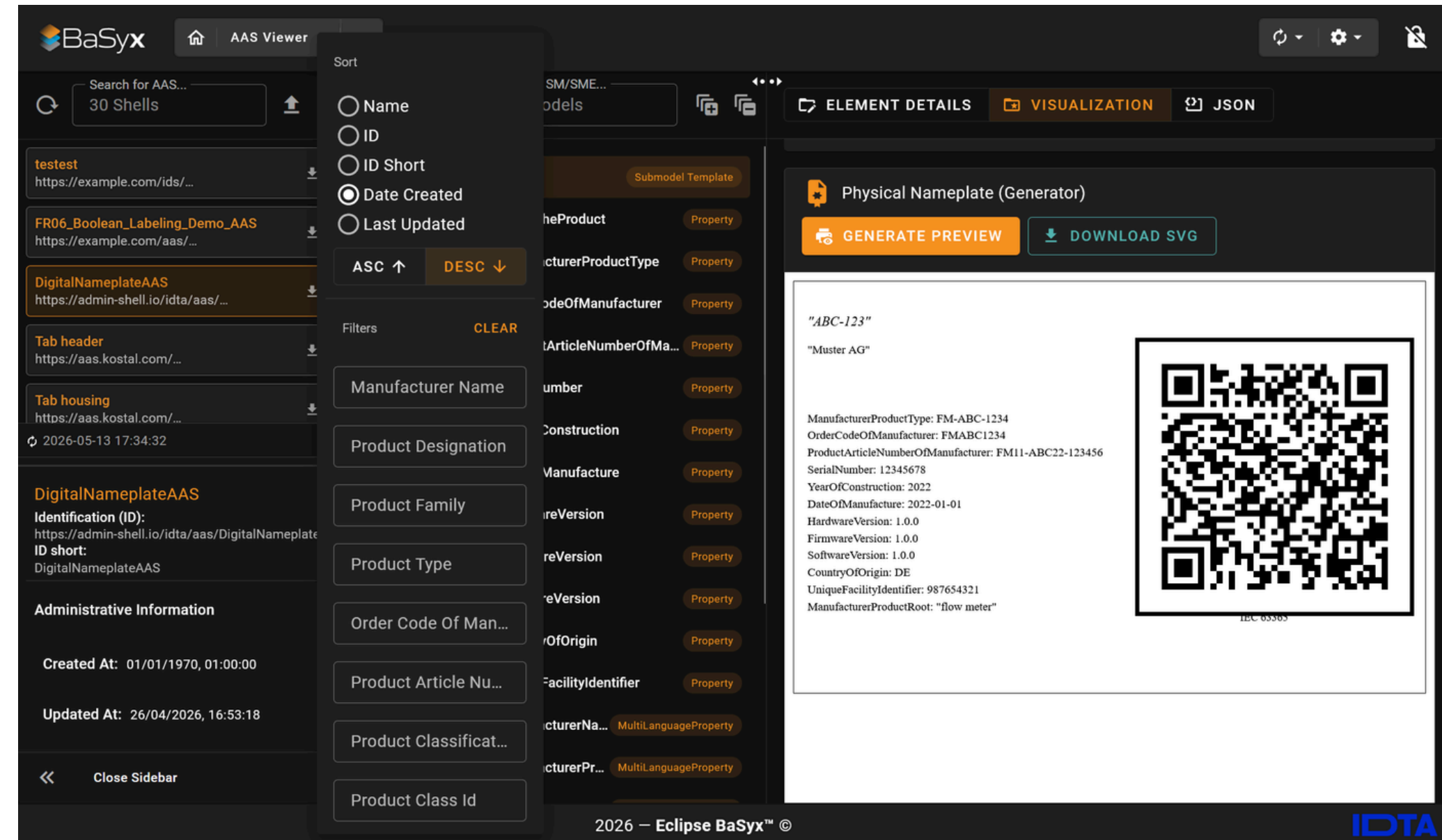


Fig. 1: Overview of BaSyx Viewer

REQUIREMENTS



SORT AAS BY PROPERTIES

- ☒ FR02
- ☐ FR03
- ☐ FR04
- ☐ FR05
- ☐ FR06

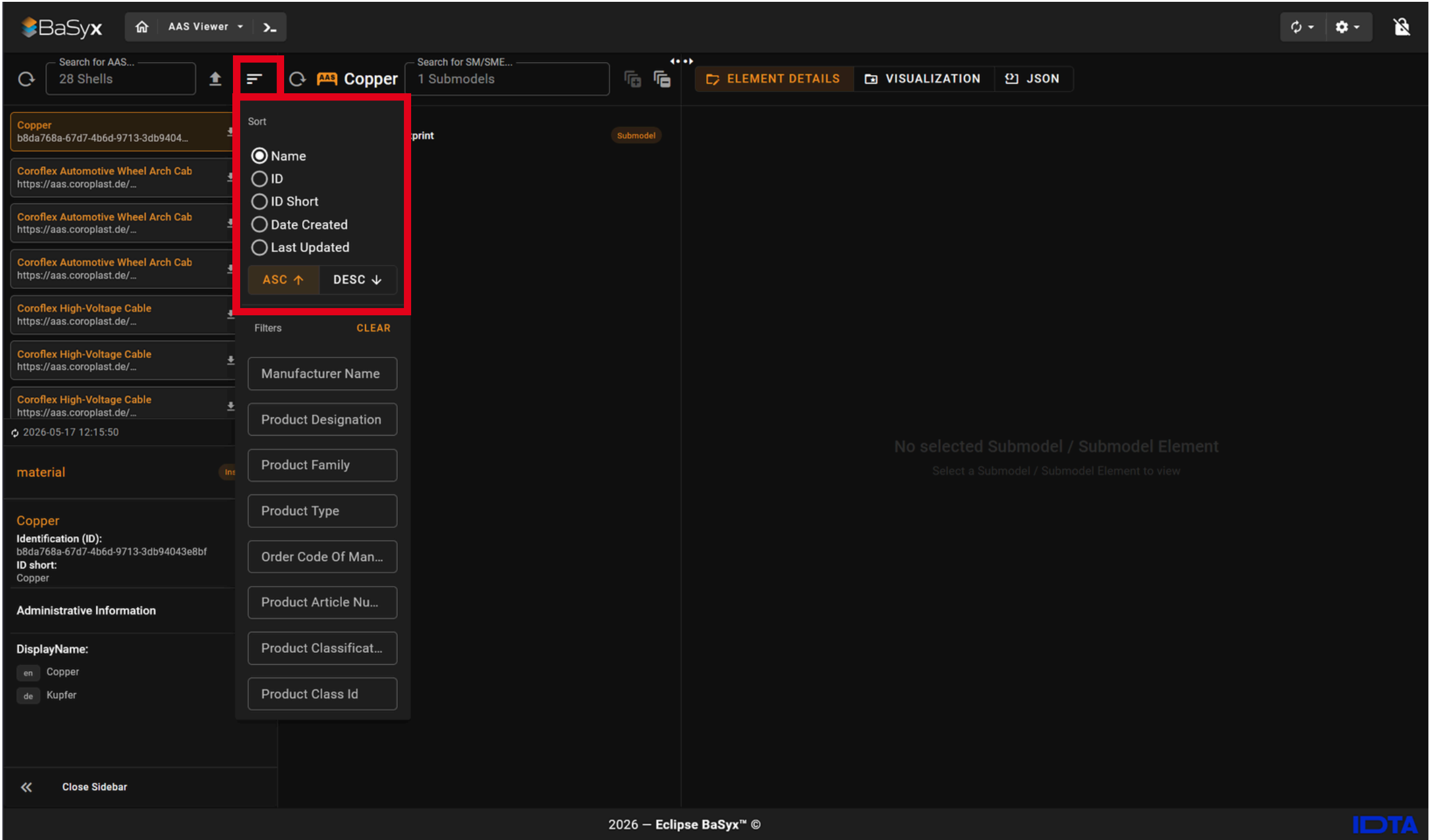


Fig. 2: Overview of BaSyx Viewer Sort

REQUIREMENTS



SORT AAS BY PROPERTIES

SORT MENU

AAS LIST

- ☒ FR02
- ☐ FR03
- ☐ FR04
- ☐ FR05
- ☐ FR06

Search for AAS...	
25 Shells	
Coroflex Automotive Wheel Arch Cab https://aas.coroplast.de/...	↓ ×
Coroflex Automotive Wheel Arch Cab https://aas.coroplast.de/...	↓ ×
Coroflex Automotive Wheel Arch Cab https://aas.coroplast.de/...	↓ ×
Coroflex High-Voltage Cable https://aas.coroplast.de/...	↓ ×

Sort

- ☒ Name
- ☐ ID
- ☐ ID Short
- ☐ Date Created
- ☐ Last Updated

ASC ↑

DESC ↓

Fig. 4: Excerpt of sorting menu

Fig. 3: List of AAS shells in viewer

REQUIREMENTS



ENHANCED SEARCH FUNCTIONALITY

- FR02
- FR03
- FR04
- FR05
- FR06

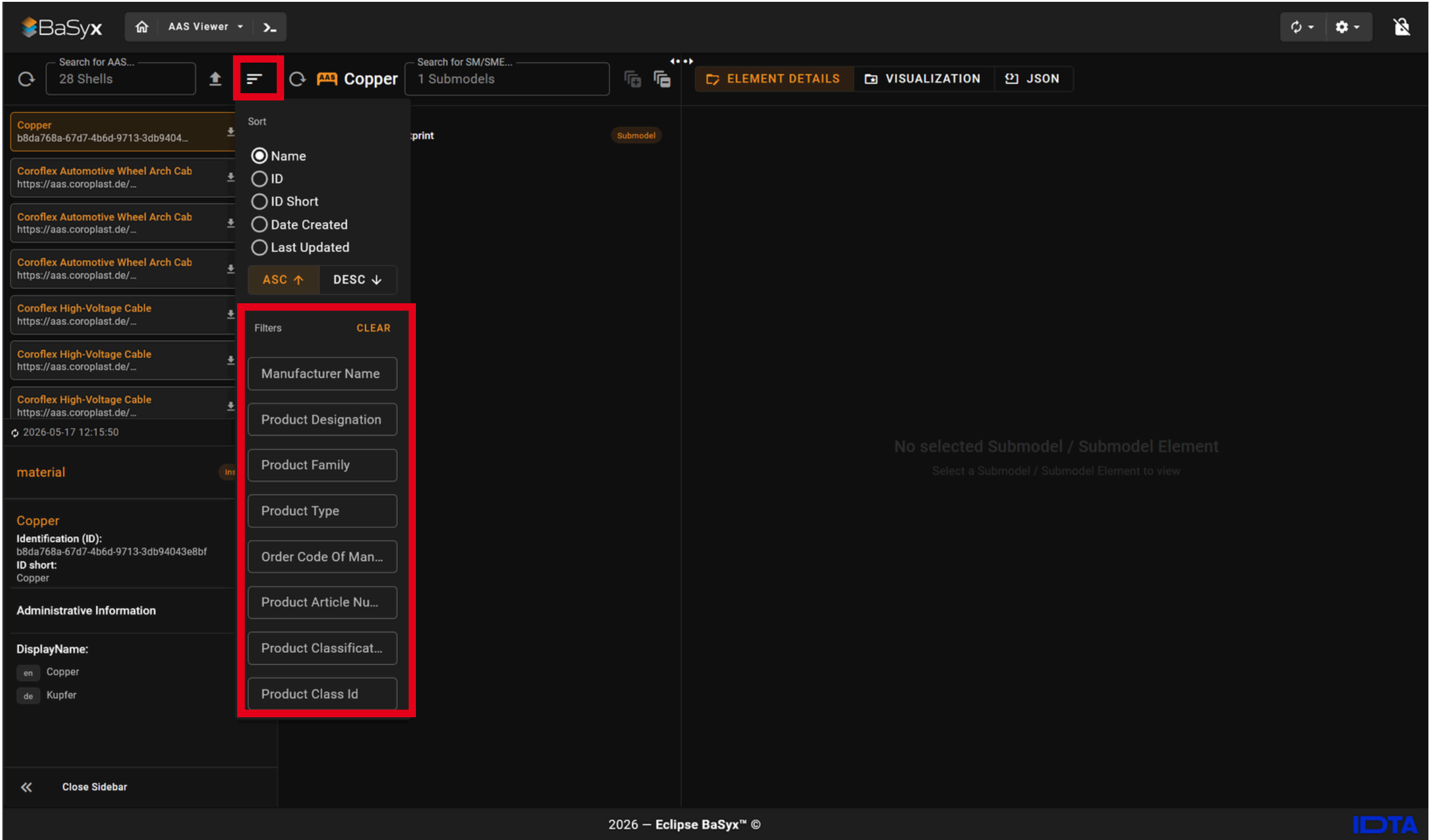


Fig. 5: Overview of BaSyx Viewer Search

REQUIREMENTS



ENHANCED SEARCH FUNCTIONALITY

- ☐ FR02
- ☒ FR03
- ☐ FR04
- ☐ FR05
- ☐ FR06

Definition Layer

aas4j

Logic Layer

basyx-java-server-sdk

uses

Presentation Layer

basyx-aas-web-ui

Rest HTTP calls

REQUIREMENTS



ENHANCED SEARCH FUNCTIONALITY

AAS LIST

- ☐ FR02
- ☒ FR03
- ☐ FR04
- ☐ FR05
- ☐ FR06

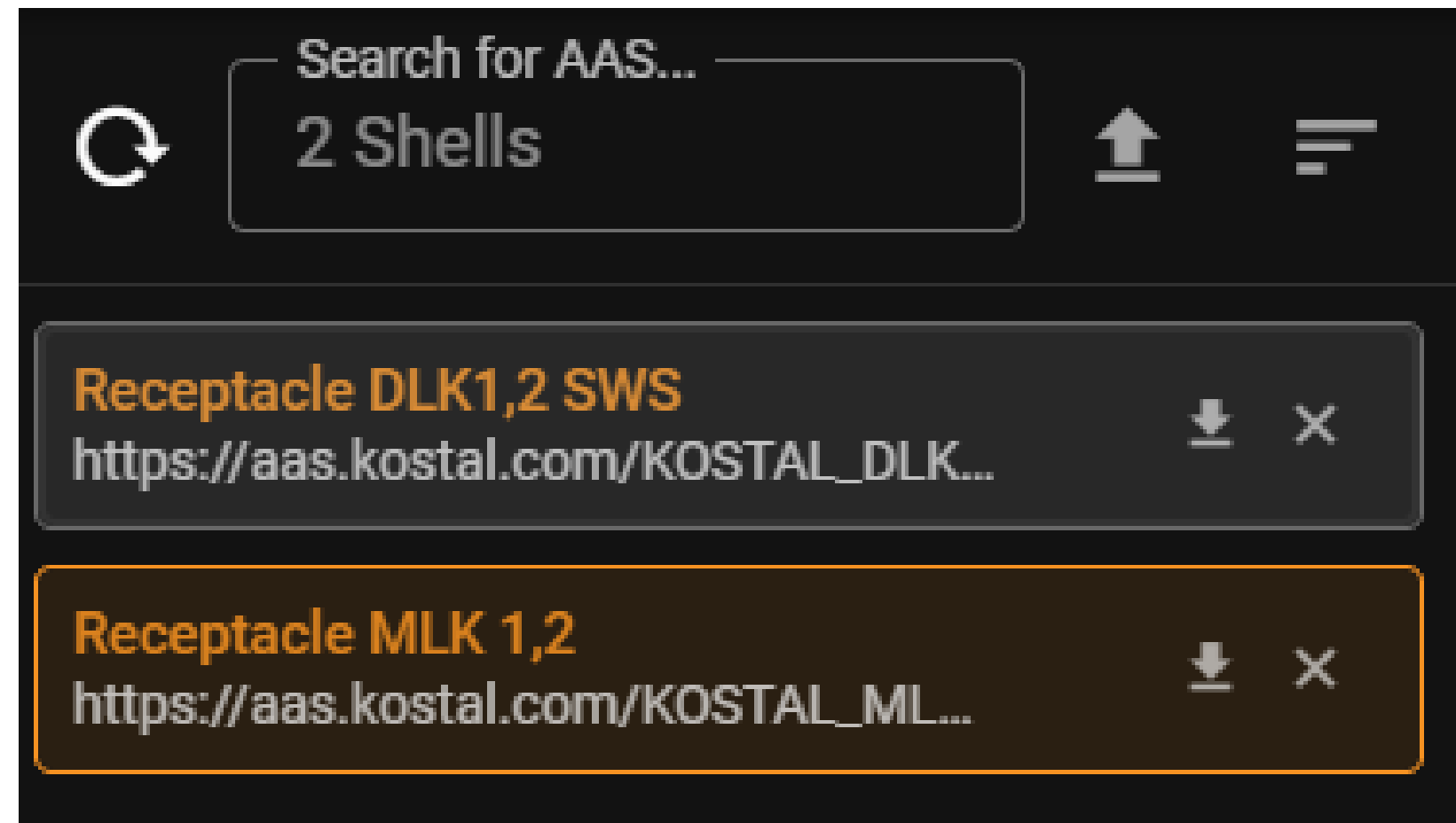


Fig. 6: List of AAS shells in viewer

FILTER MENU

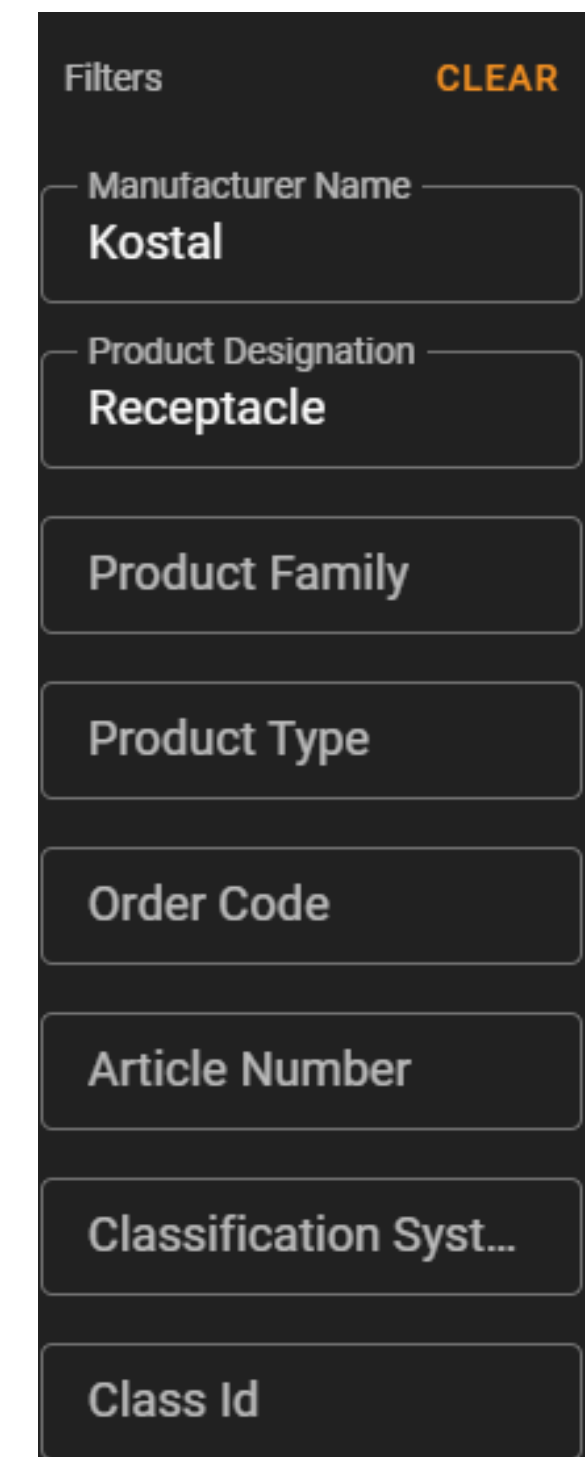


Fig. 7: Excerpt of filtering menu

REQUIREMENTS



INTEGRATE NAMEPLATE GENERATOR

- FR02
- FR03
- FR04
- FR05
- FR06

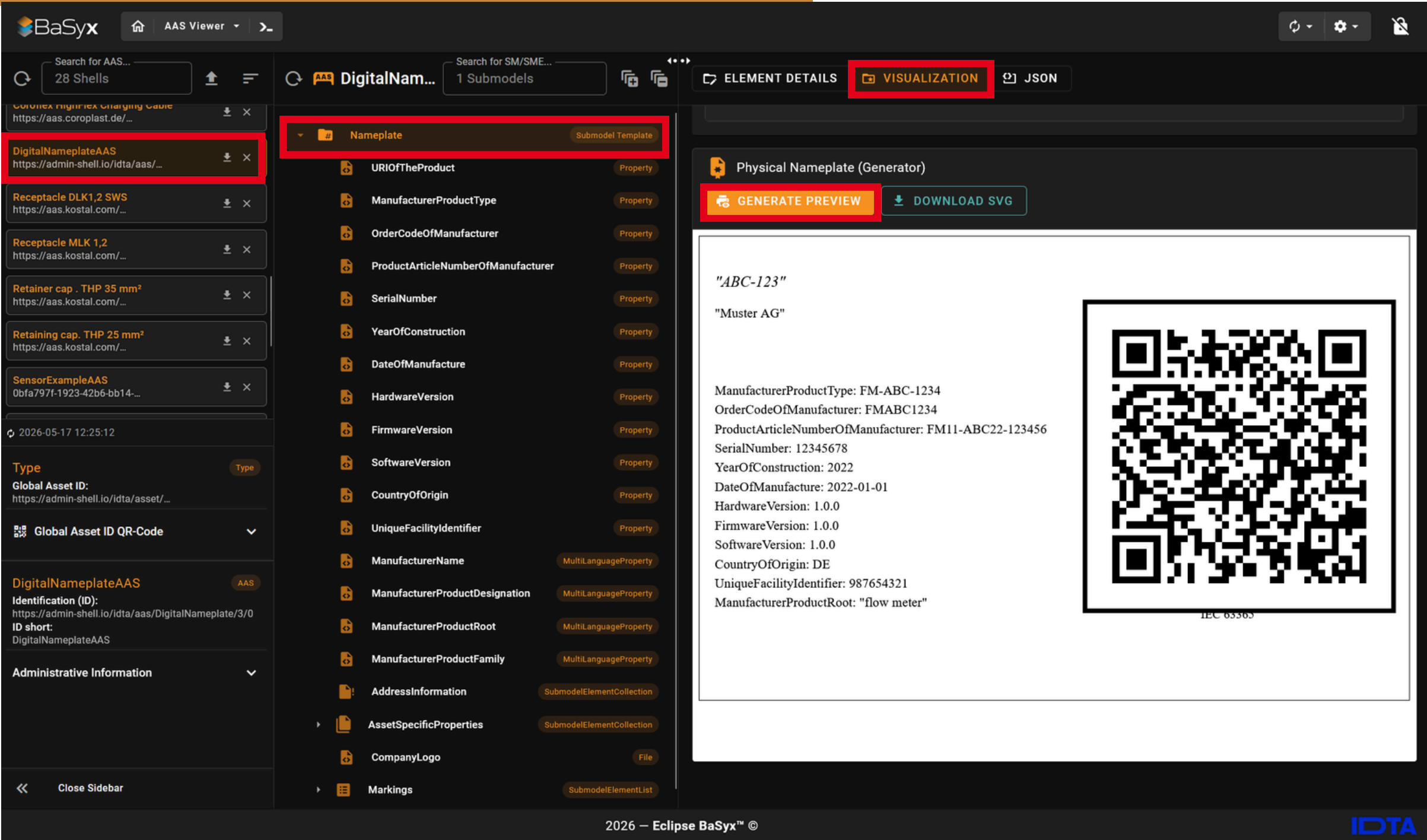


Fig. 8: Overview of BaSyx Viewer Nameplate

REQUIREMENTS



INTEGRATE NAMEPLATE GENERATOR

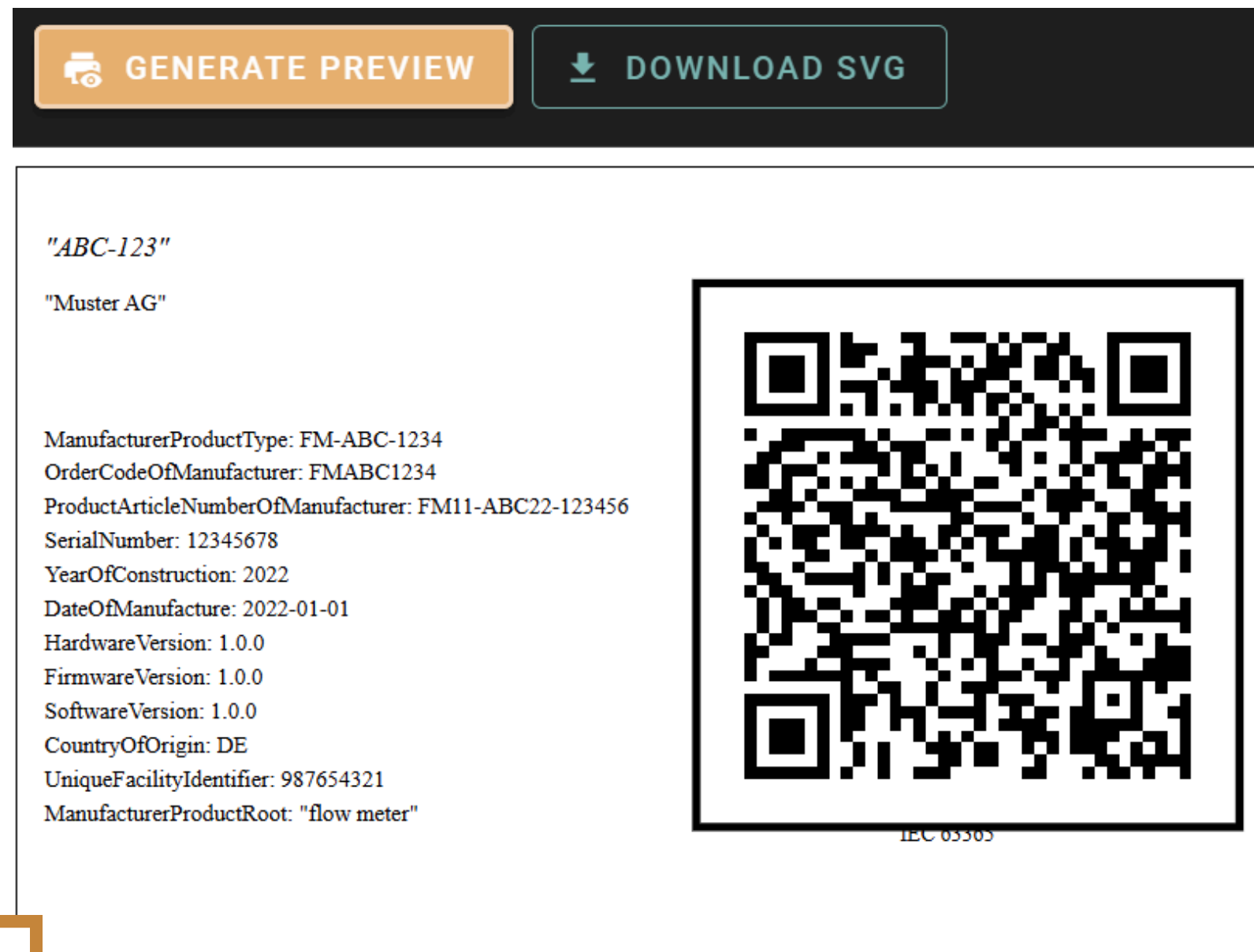


Fig. 9: Sample nameplate

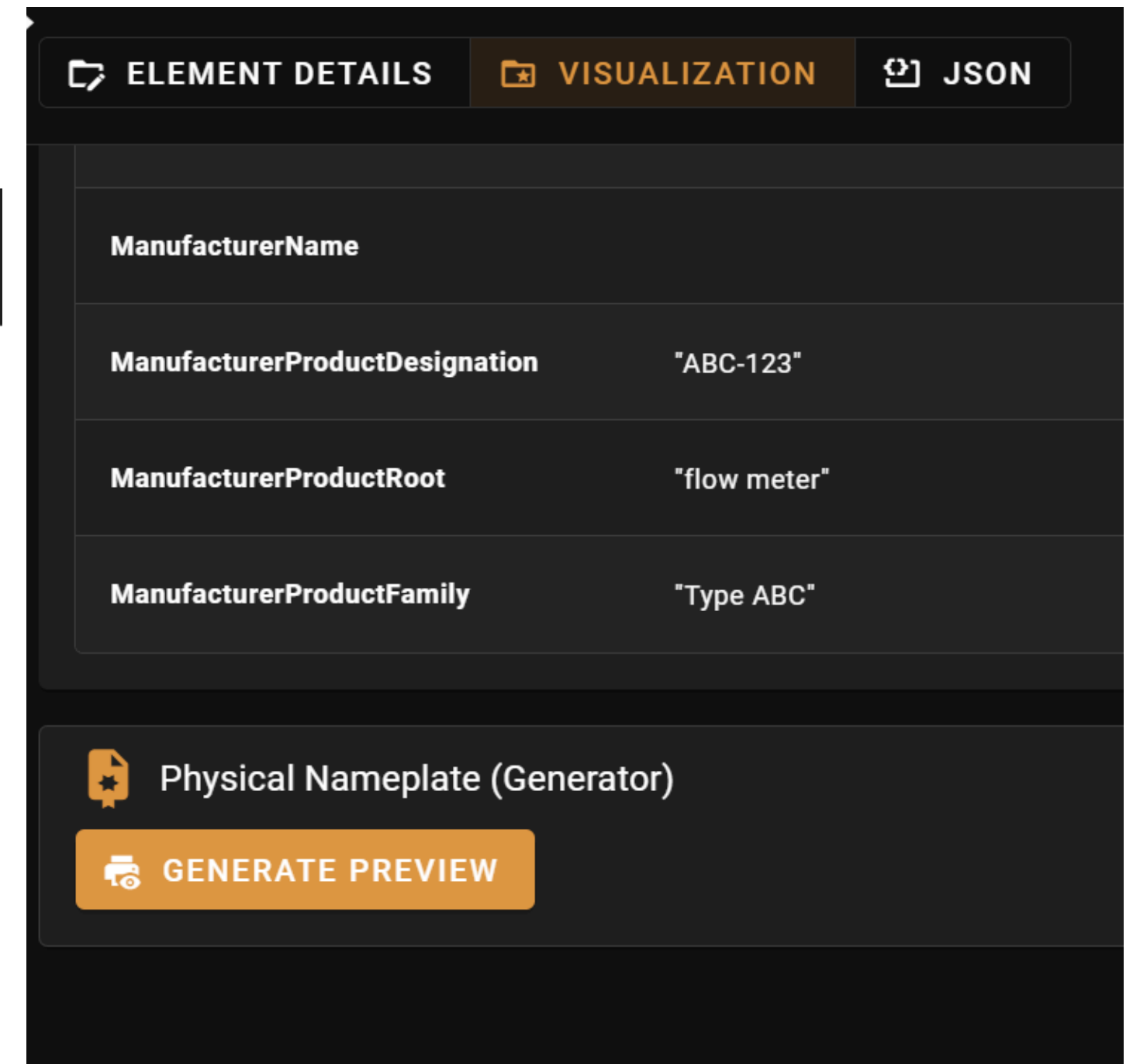


Fig. 10: User interaction for nameplate generator

REQUIREMENTS



INTEGRATE NAMEPLATE GENERATOR

NAMEPLATE3_0.VUE

FRONTEND INTEGRATION

API.JS

LOCALHOST:8080

AAS-BACKEND/NAMEPLATE-GENERATOR-BACKEND

MODIFIED CODE BY OTHER STUDENTS

Fig. 11: Implementation of nameplate generator

○ FR02

○ FR03

⦿ FR04

○ FR05

○ FR06

REQUIREMENTS



API ENHANCEMENTS

- FR02
- FR03
- FR04
- FR05
- FR06

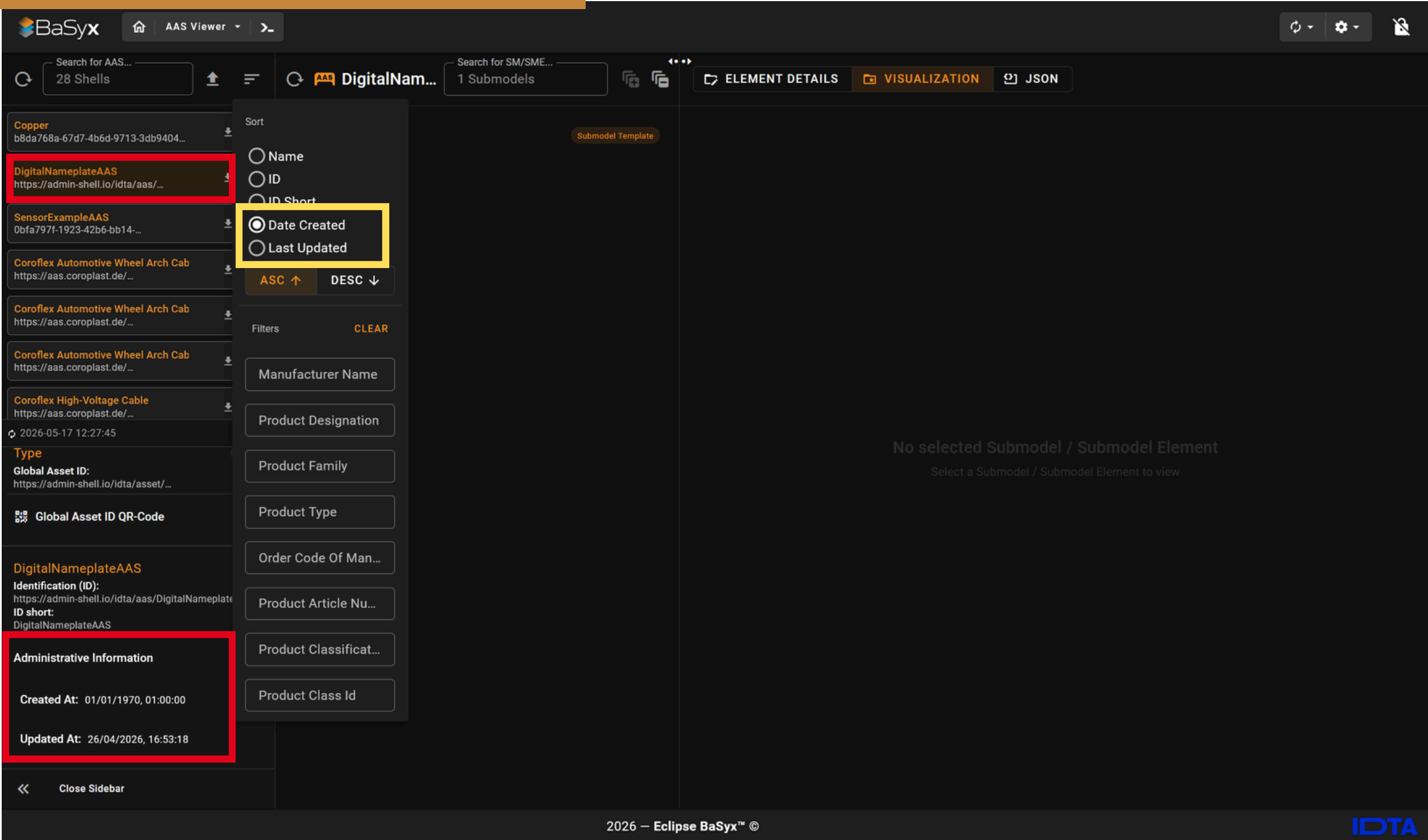


Fig. 12: Overview of BaSyx Viewer created/updated-At

REQUIREMENTS



API ENHANCEMENTS

FRONTEND INTEGRATION

Administrative Information

Created At: 01/01/1970, 01:00:00

Updated At: 02/05/2026, 15:29:32

Fig. 13: Frontend view of cA + uA params

GIVEN SPECIFICATION

Ⓒ AdministrativeInformation *HasDataSpecification*

```
+version: VersionType[0..1]
+revision:
RevisionType[0..1]
+creator: Reference[0..1]
+createdAt: dateTime[0..1]
+updatedAt: dateTime[0..1]
+templateId: Identifier[0..1]
```

Fig. 13: excerpt of IDTA aas-specifications

API RESPONSE

```
modelType: "AssetAdministrationShell"
▼ assetInformation:
  assetKind: "Instance"
  assetType: "SensorType"
▼ administration:
  version: "1.0"
  revision: "0"
  ► creator: { type: "ExternalReference",
  createdAt: "2024-03-22T09:30:00"
  updatedAt: "2024-03-22T09:30:00"
  templateId: "SensorTemplate"
  id: "aas-with-administration-001"
  idShort: "SensorExampleAAS"
```

Fig. 14: Api response from /shells

REQUIREMENTS



IMPROVE BOOLEAN LABELING OF OPERATIONS

- FR02
- FR03
- FR04
- FR05
- FR06

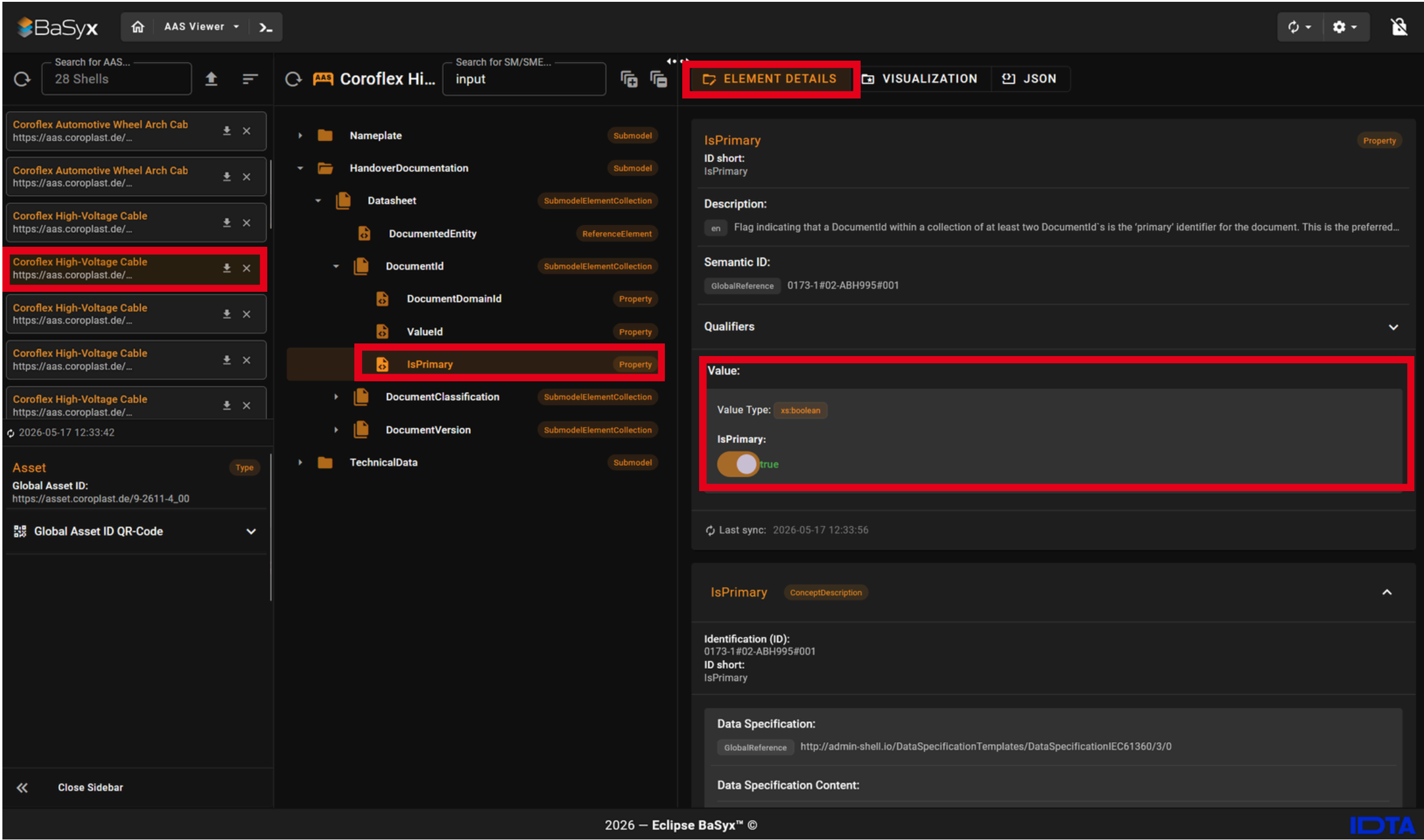


Fig. 15: Overview of BaSyx Viewer Boolean Labeling

REQUIREMENTS



IMPROVE BOOLEAN LABELING OF OPERATIONS

☐ FR02

☐ FR03

☐ FR04

☐ FR05

☒ FR06

Input Variables:

Description:

en Enables or disables the machine....

xs:boolean enableMachine



Fig. 16: boolean labeling true

Description:

en Indicates whether the emergency stop is active....

xs:boolean emergencyStop



Fig. 17: boolean labeling false

TESTS



Req. ID / Name	Functionality	Priority	Testsuit ID or TCS Link
FR.002 / Sorting (SORT)	Dynamic sorting of shells by key attributes.	A	TS.SORT.001.FAS
FR.003 / Filtering (FLTR)	Add Filters for searching shells.	A	TS.FLTR.001.FAS
FR.004 / Nameplate generator integration (GEN)	Integrate submodule into digital nameplate plugin.	B	TS.GEN.001.FAS; TS.GEN.002.FAS
FR.005 / CreatedAt and UpdatedAt support (FIELDS)	Adds two datetime fields to be recieved from the API	D	TC.FIELDS.001.F
FR.006 / Improved labeling (BOOL)	Add extra labels for Boolean values in "Operations" submodules to a visual switch instead of text.	C	TS.BOOL.001.FAS
NFR.001 / Usability	UI improvements must reduce average task time.	--	<i>no testing</i>
NFR.002 / Performance	No significant delay from enhancements.	D	TC.PERF.001.L; TC.PERF.002.L
NFR.003 / Stability	Application should not crash under standard or rapid usage.	--	<i>no testing</i>
NFR.004 / Maintainability	Clear documentation and readable code.	--	<i>no testing</i>
NFR.005 / Licensing	Maintain current open-source license.	--	<i>no testing</i>

Fig. 18: Test plan

DEMO

LIVE-LINK

DOCUMENTS/PM



○ TASK EVOLUTION

○ PROJECT MANAGEMENT

○ GITHUB STATISTICS

1. Original Task

- Search AAS List by idShort, ID and other AAS traits
- Vuetify Combobox and Filter Chips for ID-Short, ID

2. Change during Project

- Search by idShort and ID was already implemented
- Backend version limited efficient filtering for additional attributes

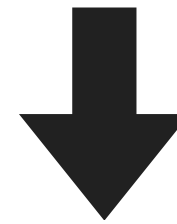
3. Final Scope

- More targeted filters for agreed attributes (Manufacturer Name, Order Code etc.)
- Filtering implemented in the frontend using backend calls

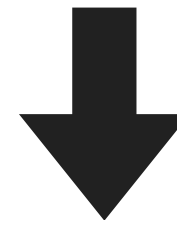
Project process



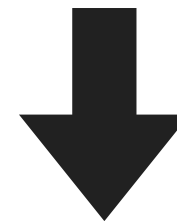
Sep–Nov 2025 Dokumentation & Planung



Mär–Apr 2026 Implementierung



Apr–Mai 2026 Tests



Mai 2026 Rollout & Abschluss

- TASK EVOLUTION
- ◎ PROJECT MANAGEMENT
- GITHUB STATISTICS

DOCUMENTS/PM

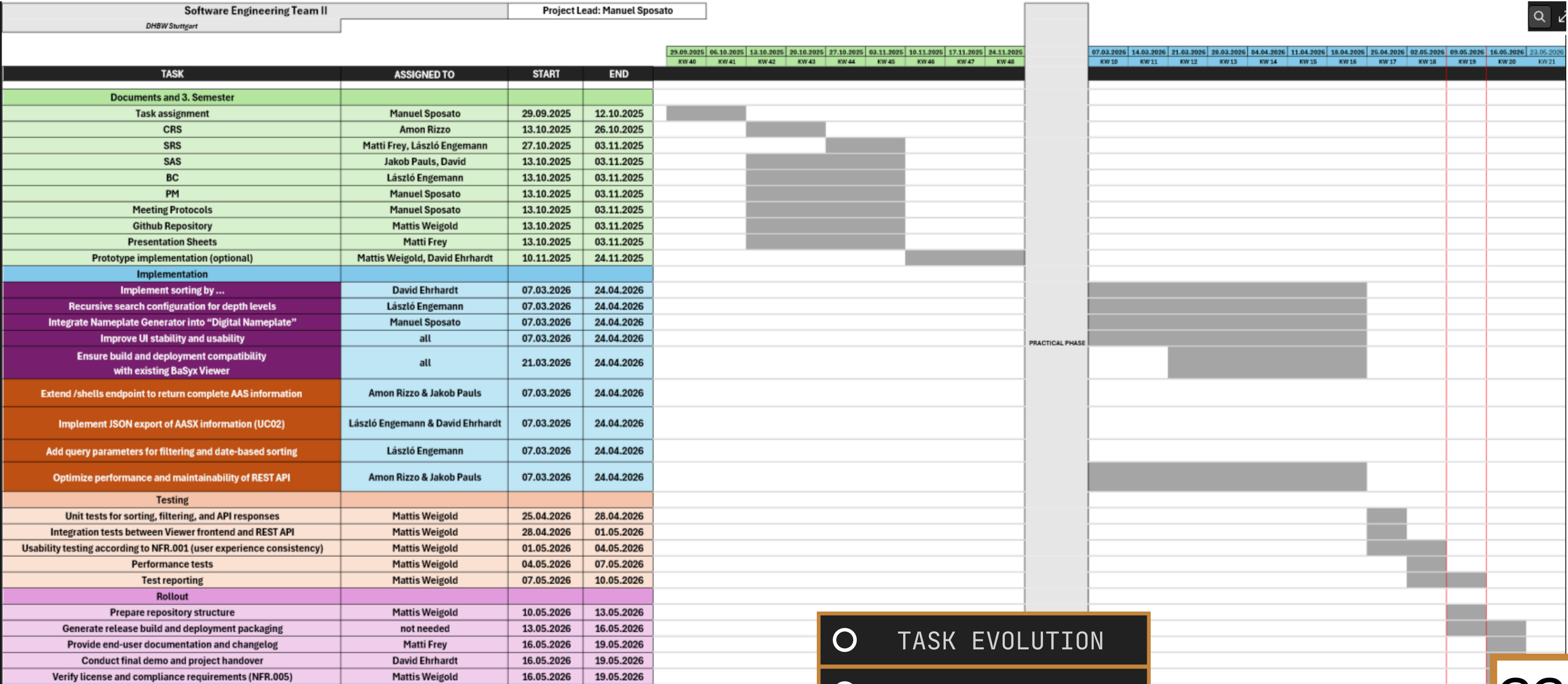


Fig. 19: Gantt diagram

- TASK EVOLUTION
- ◎ PROJECT MANAGEMENT
- GITHUB STATISTICS

DOCUMENTS/PM



- Tasks were organized by project status
- Clear overview of open, ongoing, and completed work
- Feature tasks were assigned to responsible team members
- Development, testing, documentation, and presentation tasks were tracked
- Most feature implementations were already completed
- Remaining work focused mainly on documentation and presentation
- The board helped coordinate responsibilities within the team

- TASK EVOLUTION
- ◎ PROJECT MANAGEMENT
- GITHUB STATISTICS

Fig. 20:
WhatsApp
logo



WHATSAPP + NOTION

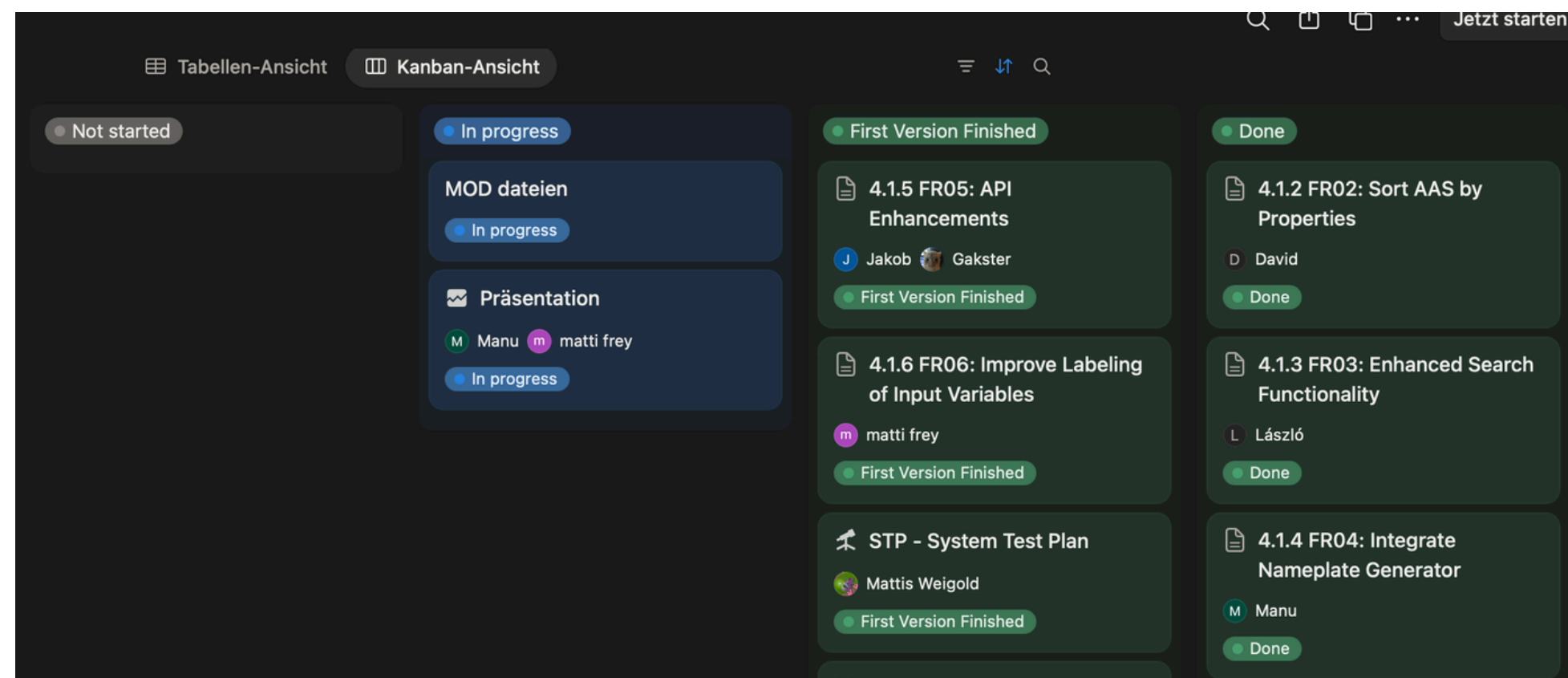
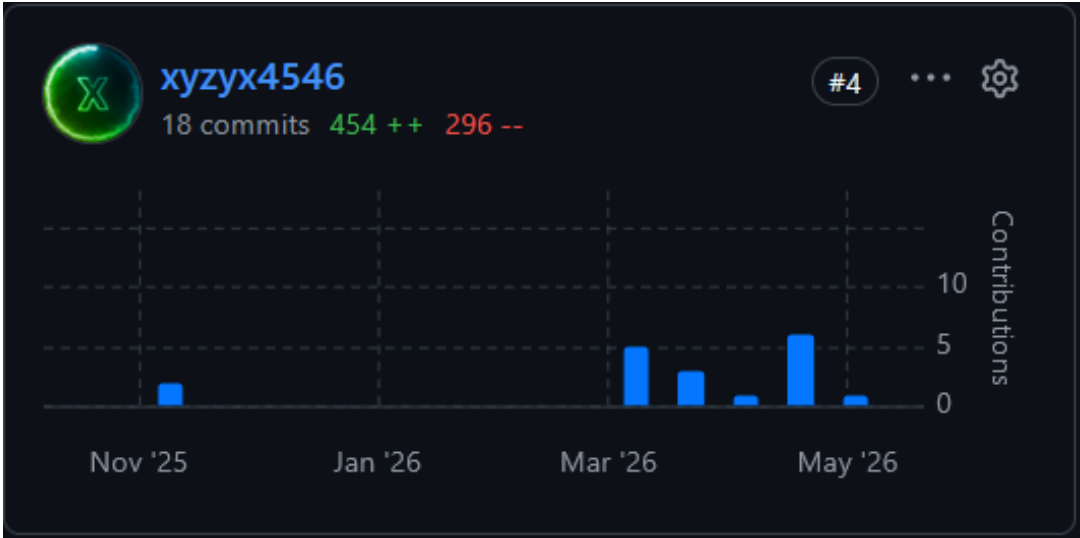


Fig. 21: Kanban board

DOCUMENTS/PM



- ☐ TASK EVOLUTION
- ☐ PROJECT MANAGEMENT
- ☒ GITHUB STATISTICS



Fig. 22: Contributions per GitHub user

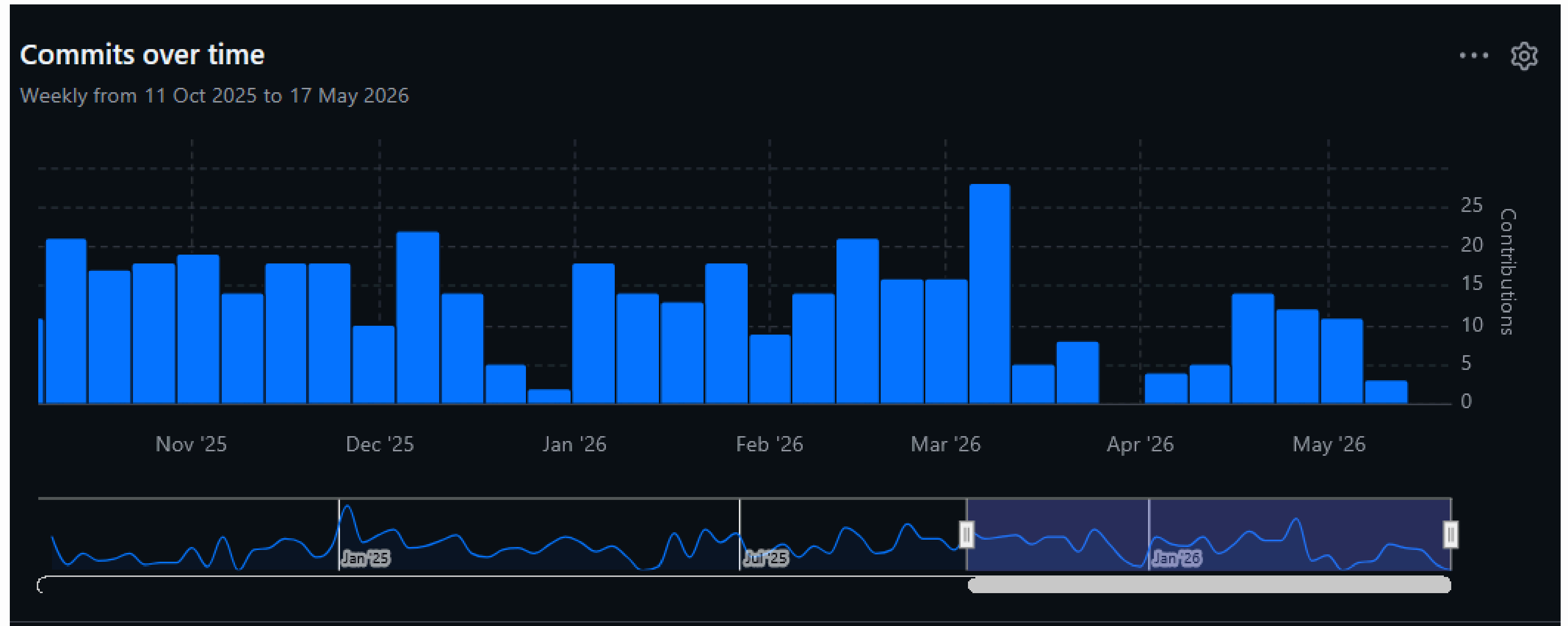


Fig. 23: Commits over time

WHAT COULD BE BETTER

CLEAR REQUIREMENTS

EARLY
SYSTEM
UNDERSTANDING

COMMUNICATION

WHAT WENT WELL

COMMUNICATION

CONTINUOUS
DOCUMENTATION

CLEAR TEAM ROLES

LESSONS LEARNED